

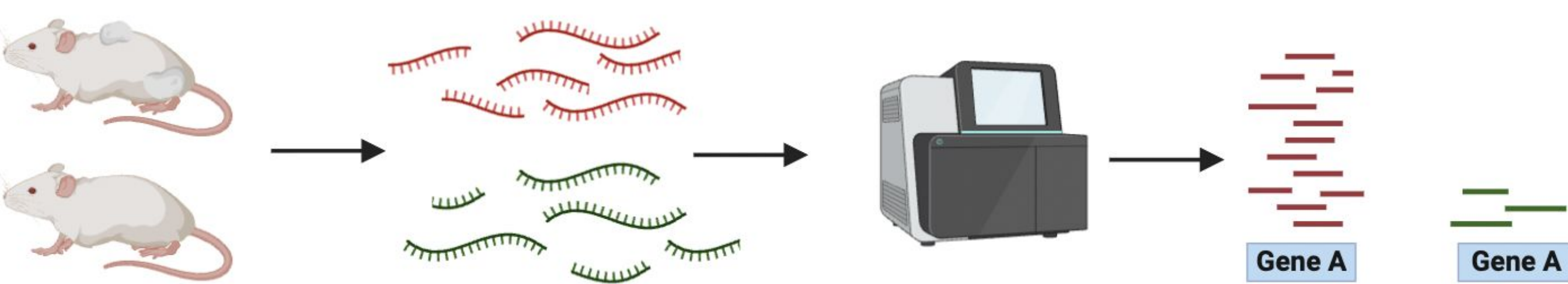
Modeling RNA-seq Count Data: A Simulation Study of Frequentist and Bayesian Approaches



GitHub Repository

Background

RNA sequencing (RNA-seq) quantifies gene expression by counting sequencing reads aligned to each gene in a sample. The resulting data are non-negative integer counts.



- Accurate statistical modeling of these counts is crucial for detecting differential expression and other downstream analyses.
- RNA-seq counts typically exhibit overdispersion (variance exceeds the mean) due to biological and technical variability.
 - The Negative Binomial distribution is often used to accommodate this extra variance via a dispersion parameter.
- The choice of likelihood can significantly impact the conclusions drawn from the data

Research Question and Objectives

We investigated how the choice of likelihood (Gaussian vs. Negative Binomial) and inferential approach (Frequentist vs. Bayesian) affects parameter estimation and uncertainty quantification for RNA-seq gene counts.

Research question: **Which statistical modeling framework provides the most accurate and reliable inference for RNA-seq count data?**

Objectives

- Simulate RNA-seq count data with known ground truth
- Fit models under four frameworks
- Estimate differential expression parameters across models
- Evaluate model performance using power and error metrics

Simulated Dataset

Simulated RNA-seq data were used in place of real datasets to allow for controlled benchmarking, with a known ground truth defining truly differentially expressed genes.

We **simulated 100 datasets** from a Negative Binomial Distribution.

10% of the genes are randomly selected to be differentially expressed.

Example count matrix

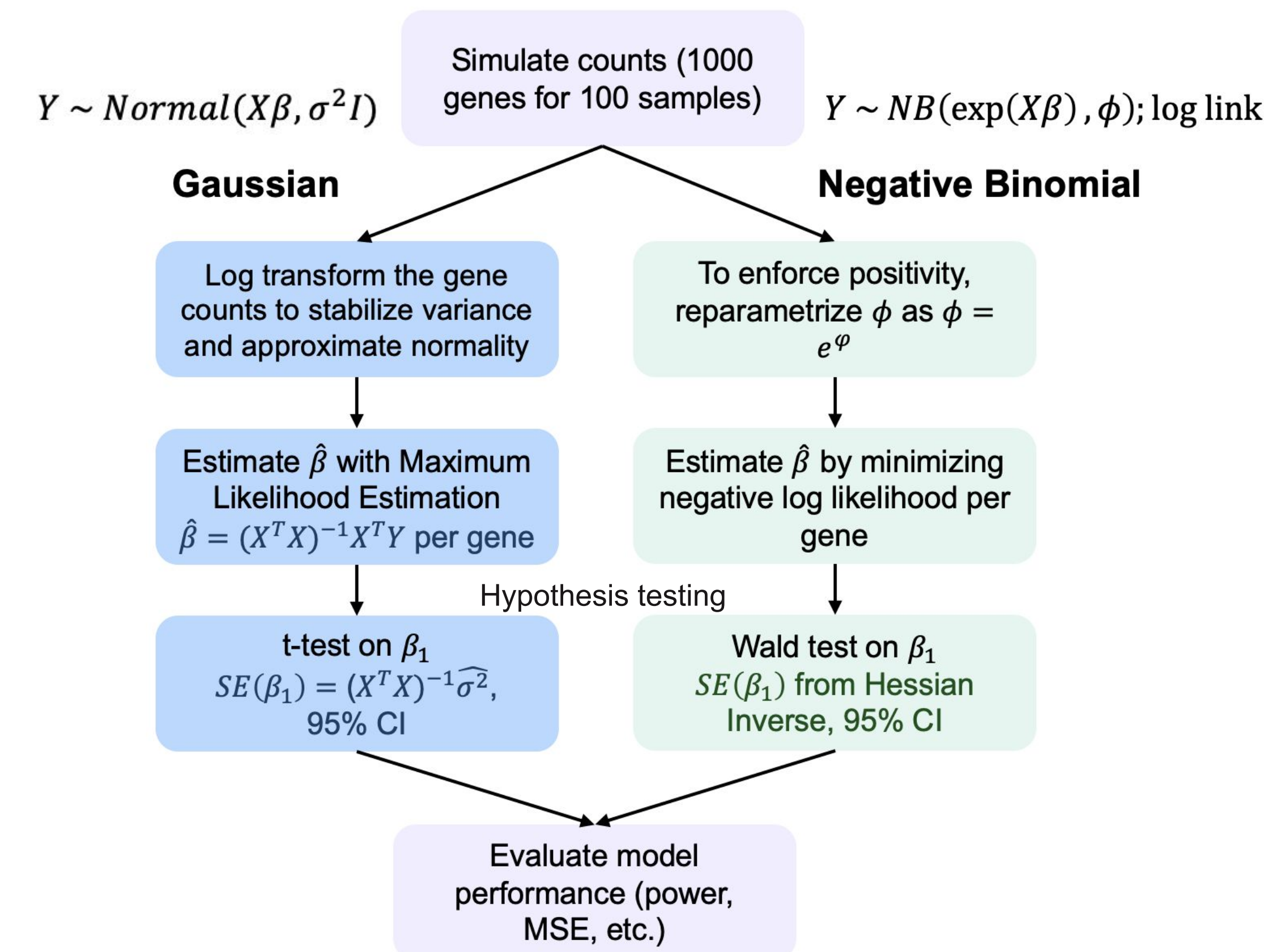
	Gene_1	Gene_2	Gene_3
Sample_1	7	3	1
Sample_2	6	3	10
Sample_3	4	3	2

Corresponding design matrix

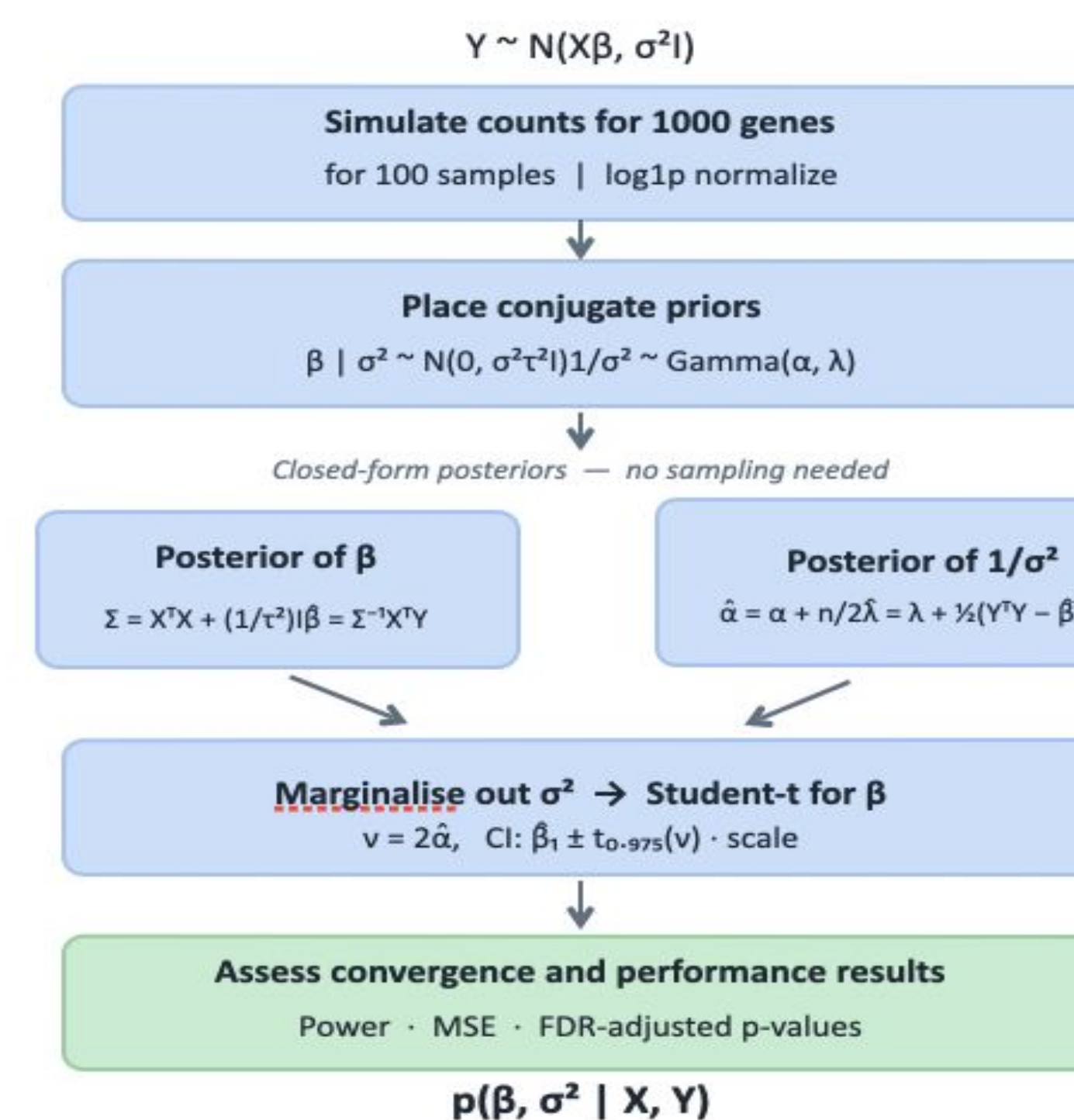
	Intercept	Group
Sample_1	1	0
Sample_2	1	0
Sample_3	1	1

Methods and Models

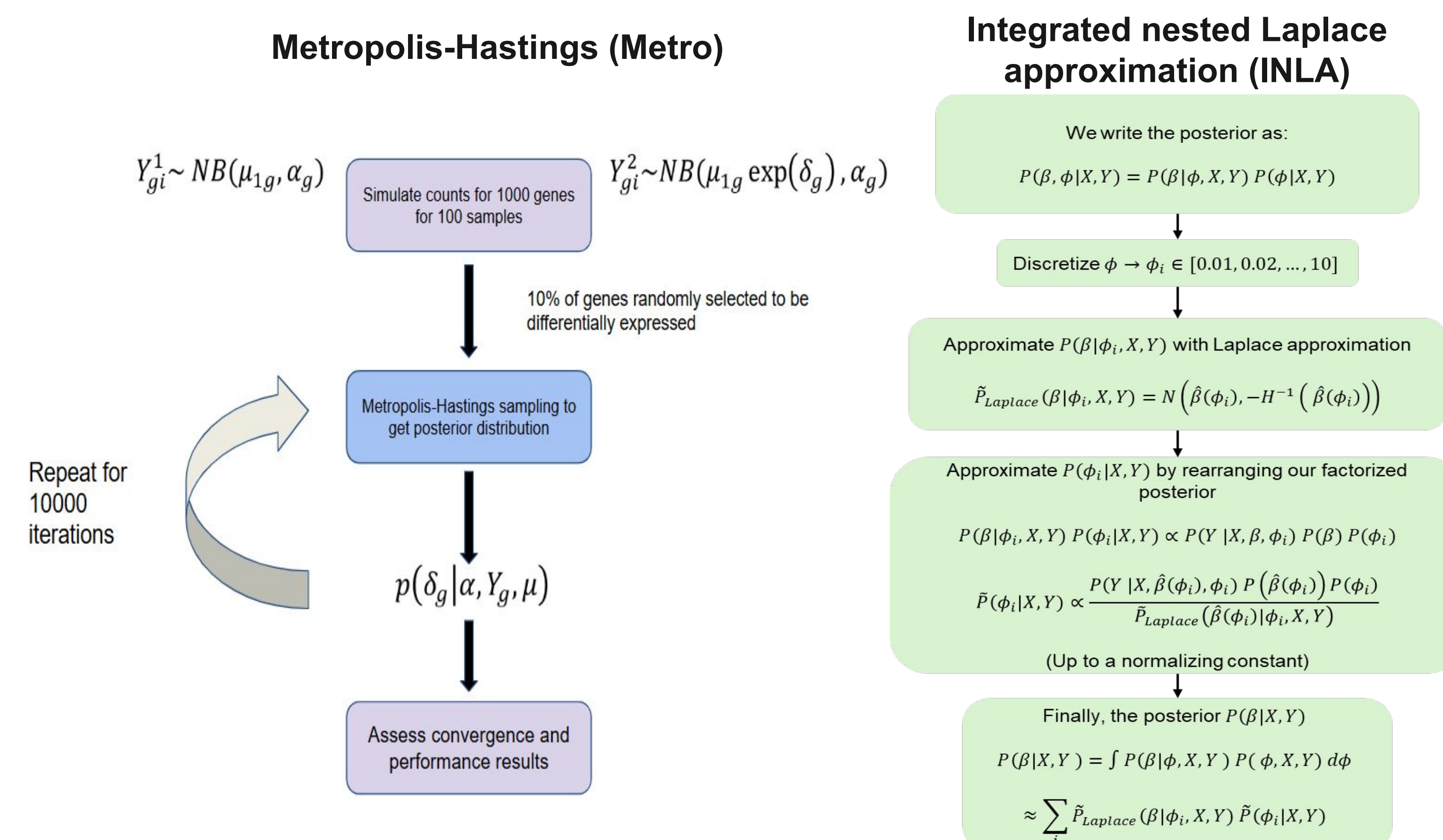
Frequentist: Gaussian and Negative Binomial



Bayesian: Gaussian

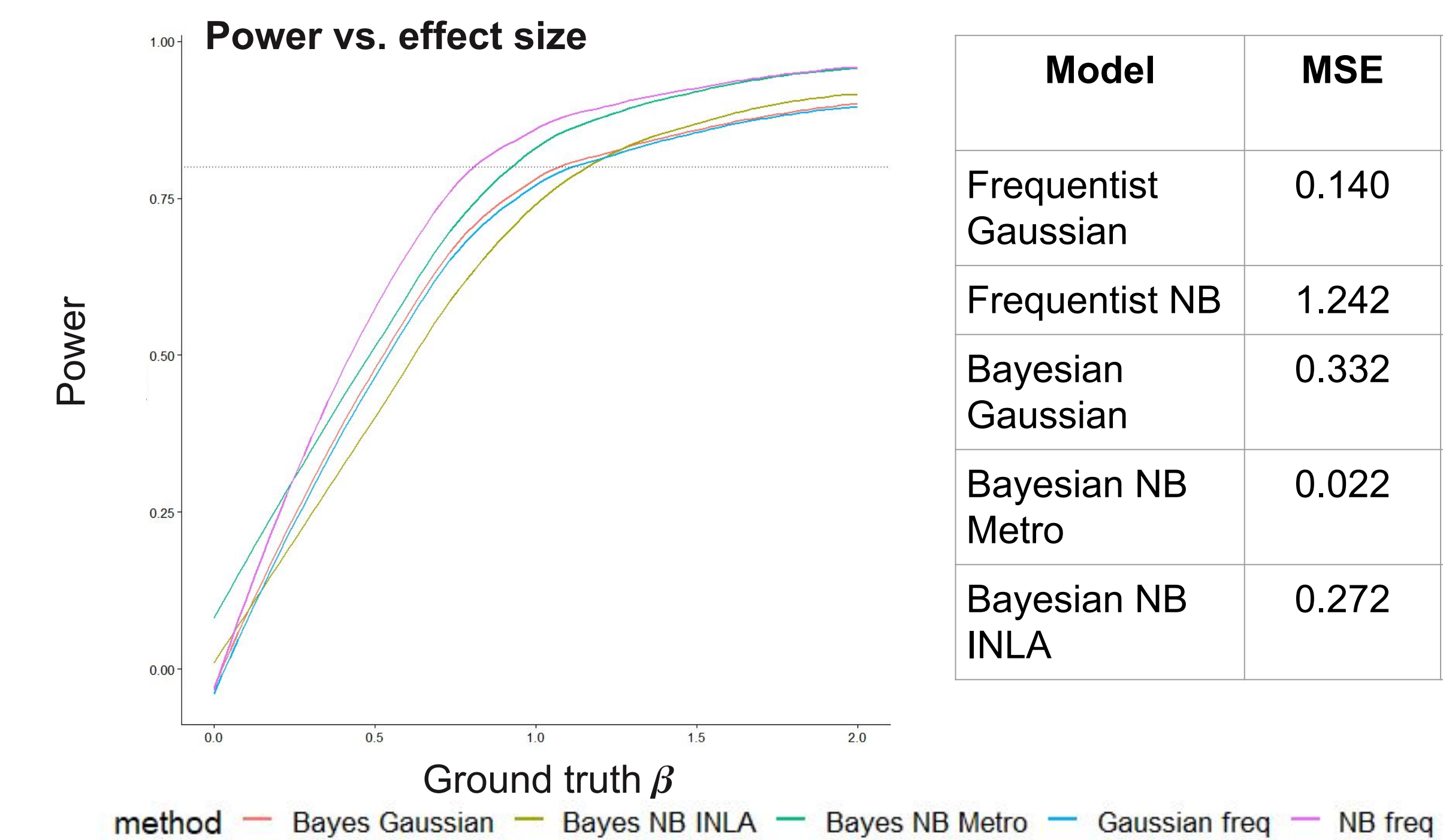


Bayesian: Negative Binomial



Results

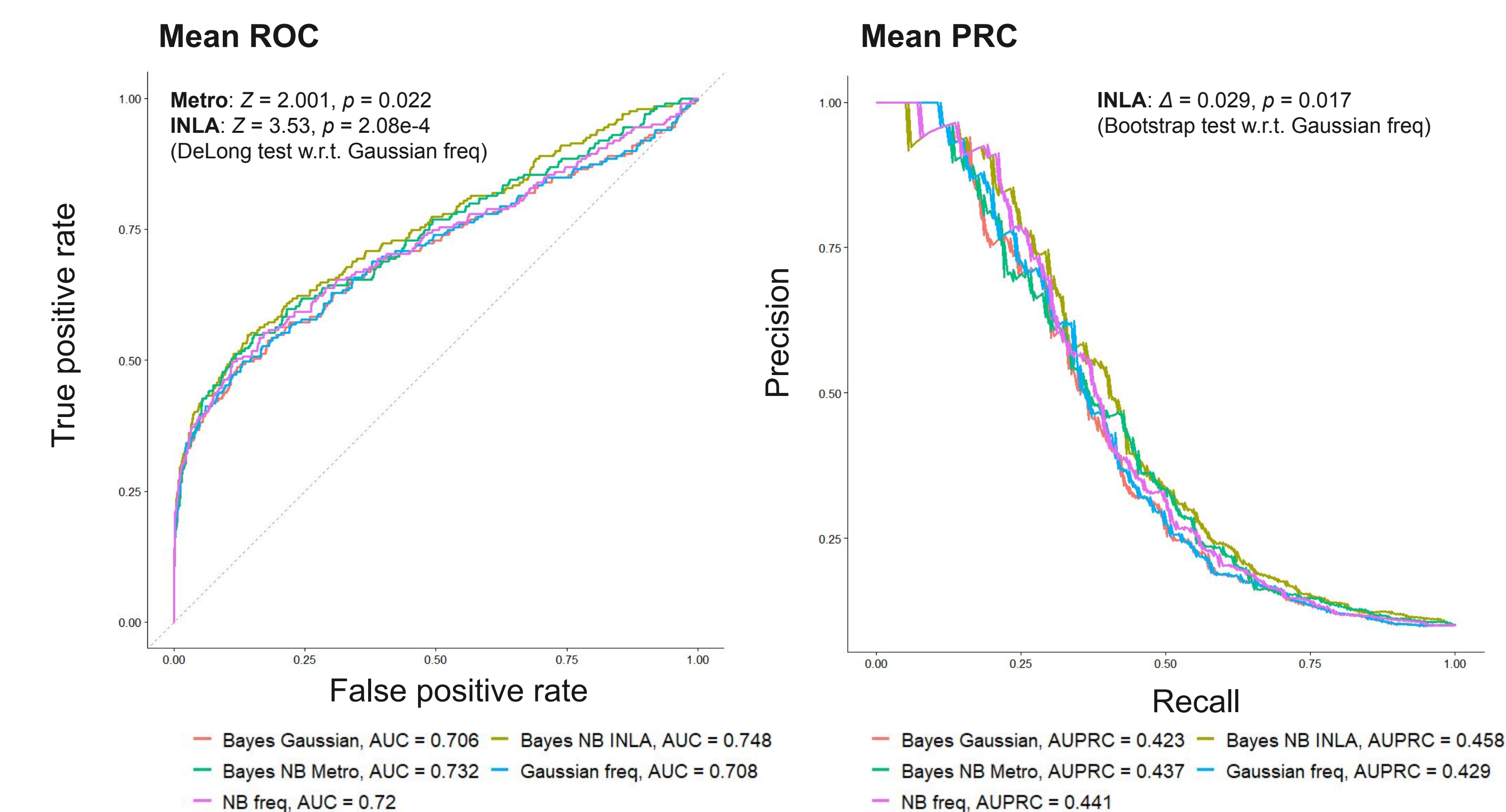
Power analysis



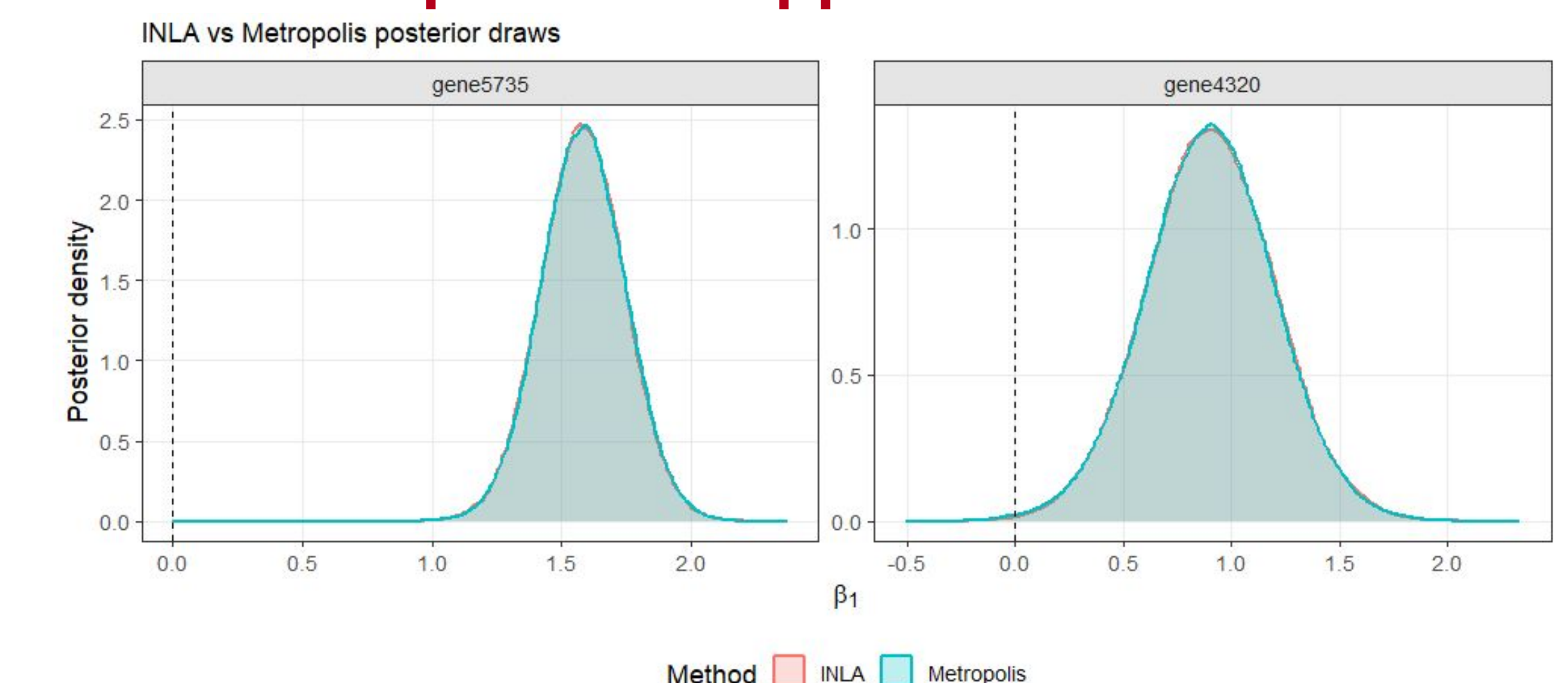
Performance Summary

Model	MSE	Type I Error	Type II Error
Frequentist Gaussian	0.140	0.0436	0.6428
Frequentist NB	1.242	0.0600	0.6022
Bayesian Gaussian	0.332	0.056	0.6562
Bayesian NB Metro	0.022	0.054	0.011
Bayesian NB INLA	0.272	0.056	0.59

AUC Analysis



Metro vs. INLA posterior approximations



Conclusions and Discussion

- Overall, the negative binomial models show improved performance over log-transformed Gaussian regression.
- The negative binomial likelihood is numerically unstable due to fold changes being sensitive to sparse counts. Marginal gain in performance when using Bayesian methods. Main benefit comes from better numerical behavior.
- We can view the Gaussian prior on the beta coefficients as restricting our search space to reasonable values, unless evidence strongly suggests otherwise.
- Frequentist negative binomial is most powerful due to lack of shrinkage on beta coefficients
- Surprisingly, Gaussian models accurately approximate fold changes. Again, the Bayesian version provides numerical stability, as log fold changes are unstable when counts are sparse.

Citations

Anders, S., Huber, W. Differential expression analysis for sequence count data. Genome Biol 11, R106 (2010). <https://doi.org/10.1186/gb-2010-11-10-r106>